



Evaluation of the GCdiaMTB/NTM real-time PCR kit according to the Korean *in vitro* diagnostic medical device approval standards[☆]

Seo Yong Choe^a, Da-Gyum Lee^a, Jin-A Park^{a,b,1}, Munki Yoo^{a,2}, Jaehyeon Kang^{a,b}, Tak Gyeom Lee^{a,3}, JeongA Kim^{a,1}, Dabin Jeong^{a,4}, Soozy Kim^a, Younghoon Jang^b, Heedoo Lee^b, Jichan Jang^c, Sungweon Ryoo^{a,b,*,1}

^a Clinical Research Centre, Masan National Tuberculosis Hospital, Changwon 51755, Republic of Korea

^b Department of AI-Integrated Biological Sciences, GAST (Global Advanced Institute of Science & Technology), Changwon National University, Changwon 51140, Republic of Korea

^c Division of Life Science, Department of Bio & Medical Big Data (BK21 Four Program), Research Institute of Life Science, Gyeongsang National University, Jinju 52828, Republic of Korea

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ABSTRACT

Background: *Mycobacterium tuberculosis* complex (MTBC) and non-tuberculous mycobacterial (NTM) infections present with similar clinical symptoms but require different treatments, so accurate, rapid differential diagnosis is essential. While traditional methods, such as acid-fast bacilli (AFB) microscopy and culture, are time-consuming, molecular diagnostics offer a faster alternative.

Methods and results: This study evaluated the clinical performance of the GCdiaMTB/NTM Real-Time PCR Kit (GCdia) for diagnosing MTBC and NTM infections. The kit was compared to a licensed product, the AdvansureTB/NTM real-time PCR kit (Advansure), using 660 residual sputum and culture specimens from patients suspected of having pulmonary tuberculosis (TB) and NTM infections, classified using a composite reference standard (AFB culture, with the MPT64 antigen test used to differentiate MTBC from NTM in positive cultures). The kit's performance was evaluated in accordance with the Korean Ministry of Food and Drug Safety (MFDS) *in vitro* diagnostic medical device (IVD) approval standards. For sputum specimens, GCdia showed 100% sensitivity and 99.32% specificity for MTBC, and 100% sensitivity and 100% specificity for NTM (95% CI reported for all estimates). In a paired comparison using the exact McNemar test, GCdia showed a statistically significant advantage over Advansure for NTM sensitivity in sputum (10.77 percentage points; 95% CI 3.23–18.31; $p = 0.016$); differences in MTBC sensitivity in sputum and in both targets in culture specimens were not statistically significant. Agreement between GCdia and Advansure was excellent across all specimen and target categories (Cohen's kappa 0.921–1.000).

Conclusions: GCdia showed clinical performance comparable to, or in one comparison significantly higher than, Advansure, and met all predefined Korean MFDS regulatory thresholds. Because the reference standard used for NTM classification does not include species-level confirmation, interpret these findings as applying to broad

Abbreviations: Advansure, Advansure TB/NTM real-time PCR kit; AFB, Acid Fast Bacilli; CI, confidence interval; DNA, Deoxyribonucleic Acid; FAM, Fluorescein Amidite; FDA, Food and Drug Administration; GCdia, GCdia MTB/NTM Real-Time PCR Kit; HEX, Hexachlorofluorescein; IC, Internal Control; IGRA, Interferon Gamma Release Assay; IRB, Institutional Review Board; IVD, In Vitro Diagnostic; MDR, Medical Device Regulation; MDR-TB, Multidrug-resistant tuberculosis; MFDS, Ministry of Food and Drug Safety; MGIT, Mycobacterial Growth Indicator Tube; MNTH, Masan National Tuberculosis Hospital; MTB, *Mycobacterium tuberculosis*; NPA, Negative Percent Agreement; NTM, Non-tuberculous Mycobacteria; OPA, Overall Percent Agreement; PCR, Polymerase Chain Reaction; PMA, Premarket Approval; PPA, Positive Percent Agreement; SOP, Standard Operating Procedure; TB, Tuberculosis; WHO, World Health Organization.

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* Corresponding author at: Clinical Research Centre, Masan National Tuberculosis Hospital, Changwon 51755, Republic of Korea.

E-mail addresses: viweon@changwon.ac.kr, viweon@mycodx.com (S. Ryoo).

¹ Present Address: Biotechnology R&D Center, MycoDx Co., Ltd., Changwon 51139, Republic of Korea.

² Present Address: Forensic Medicine Division at the National Forensic Service Busan Institute, Yangsan 50612, Republic of Korea.

³ Present Address: Forensic Medicine Division at the National Forensic Service Busan Institute, Yangsan 50612, Republic of Korea.

⁴ Present Address: Division of Bacterial Diseases Research, Korea National Institute of Health, Osong 28159, Republic of Korea.

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detection of NTM growth rather than species-specific diagnostic accuracy. Our findings provide a practical guide for researchers and companies preparing for Korean IVD approval.

1. Introduction

Tuberculosis (TB), caused by MTB, is a major public health concern worldwide, caused by the MTBC. It is an acute and chronic disease that usually affects the lungs but can occur anywhere in the body, including the lymph nodes or spinal cord. The rise of multidrug-resistant tuberculosis (MDR-TB) has made treatment even more challenging. The World Health Organization (WHO) estimates that approximately one-quarter of the world's population is infected with MTBC. In 2023, approximately 10.8 million people were diagnosed with TB, and approximately 1.25 million died from it (World Health Organization (WHO), 2024). Simultaneously, NTM, which can be the causative pathogen of infectious diseases that cause social alarm, along with TB, refers to mycobacteria other than MTB and *M. leprae* and mainly exist in natural environments such as soil and water. NTM diseases are classified into four types based on their clinical manifestations: pulmonary disease, lymphadenitis, skin disease, and disseminated disease. Pulmonary diseases account for more than 90% of cases (American Thoracic Society, 1997; Griffith, 2007).

Although NTM infections present with clinical symptoms and imaging findings similar to those of TB, their treatment and prognosis are entirely different, making an accurate differential diagnosis crucial. Traditional methods used to diagnose TB (e.g., Acid-Fast Bacillus (AFB) microscopy and culture) are time-consuming and have limitations for rapidly distinguishing MTB from NTM. Therefore, it is essential to develop a molecular diagnostic method that can rapidly and accurately distinguish between MTBC and NTM infections. This study aimed to validate the clinical performance of a novel molecular diagnostic method for early diagnosis and personalized treatment of these diseases.

In the United States, IVDs used to diagnose MTBC and NTM infections must be approved by the Food and Drug Administration (FDA). The FDA classifies IVDs into Class I, Class II, and Class III categories, based on the level of regulatory control. Previously, molecular diagnostic IVDs for TB required Premarket Approval (PMA) as a Class III device. However, the approval process was reorganized in 2013, and they were reclassified as Class II. FDA approval must now be obtained under the Class II special control guidelines (U.S. Food and Drug Administration, 2014). In 2023, there was a discussion about reclassifying the interferon gamma release assay (IGRA) kit, which aids in diagnosing latent tuberculosis, from Class III to Class II (U.S. Food and Drug Administration, 2023). Europe has implemented strengthened medical device regulations (MDR) and IVD regulations from 2021 to 2022. Europe also divides IVDs into four classes (classes A–D) according to their potential risk levels. Molecular diagnostic IVDs for TB are class C. Except for class A, the rest must undergo examination and certification by an IVD certification body (European Medicines Agency, 2017). Because of tightening European regulations, existing IVDs must be converted to comply. The transition period was extended from May 2026 to December 2028, depending on the class and type of medical device (European Medicines Agency, 2023).

In Korea, the IVD Act is classified into four grades based on safety. Since molecular diagnostic IVDs for TB are classified as grade 3, they must be approved by the Korean MFDS and have test data from a clinical performance testing institution (Ministry of Food and Drug Safety, 2020). The Clinical Research Center at Masan National TB Hospital (MNTH), which met the requirements for a clinical performance testing institution (Ministry of Food and Drug Safety, 2021), was designated as an IVD clinical performance testing institution (IVD Notice No. 79) on July 1, 2021. In 2023, the Korean MFDS developed, registered, and approved 14 molecular and/or immunodiagnostic products for TB (Ministry of Food and Drug Safety, South Korea, 2023). In 2024, GCdia

(Genes Lab, Gyeonggi-do, Korea) was tested for clinical performance at the MNTH Clinical Research Center.

In this study, we validated the GCdia method against the Advansure control kit (Advansure, Invitrogen, Osong, Korea) and against AFB microscopy and culture results. We also share our experience with the Korean MFDS approval process during clinical performance trials as a practical methodological reference for researchers and companies preparing for Korean IVD approval of molecular diagnostic methods.

The GCdia method tested in this study demonstrated sensitivity and specificity for TB/NTM detection in sputum and cultured samples that were comparable to, or in one comparison significantly higher than, the previously approved product, Advansure.

2. Materials and methods

2.1. Study design

This retrospective single-center clinical trial was conducted at the Clinical Research Center of MNTH to validate the performance of a new molecular diagnostic method. We stored specimens from participants suspected of having pulmonary TB or NTM infection from 2018 to 2024, using residual sputum and cultured specimens, at -80°C in the MNTH Biobank. We selected specimens based on AFB microscopy and culture test results, in accordance with the Korean Tuberculosis Clinical Guidelines (Korean Academy of Tuberculosis and Respiratory Diseases, Committee for the Revision of the Korean Guidelines for Tuberculosis, 2024). A total of 660 specimens were included, comprising 216 MTB-positive, 135 NTM-positive, and 309 MTB- and NTM-negative specimens. The Institutional Ethics Committee of MNTH approved the study protocol (IRB-398837MD-2024-E01).

2.2. Specimen pretreatment and reference methods

Sputum specimens were decontaminated with 2% sodium hydroxide/standard *N*-acetyl-L-cysteine (NaOH–NALC) and then dispensed into conical tubes containing 30 mL of phosphate-buffered saline (PBS, pH 7.4). The specimens were then vortexed and centrifuged at 4000 rpm for 10 min at 25°C . The supernatant was discarded. This process was repeated, and the samples were stored in a -80°C deep freezer at a final concentration of 8% glycerol (Korean Academy of Tuberculosis and Respiratory Diseases, Committee for the Revision of the Korean Guidelines for Tuberculosis, 2024).

2.3. AFB microscopy and culture

AFB microscopy was performed using Ziehl-Neelsen staining or a fluorescence microscope at $200\text{--}250\times/400\text{--}450\times$ magnification, respectively, with auramine-rhodamine fluorescence staining (World Health Organization, 2011). This study followed the AFB culture protocol described in the TB treatment guidelines (World Health Organization (WHO), 1998). We cultured all specimens simultaneously using two methods: 3% Ogawa solid medium (Asan Pharm, Seoul, Korea) and a mycobacterial growth indicator tube (MGIT, Becton, Dickinson and Company, Franklin Lakes, NJ, USA) (Korean Academy of Tuberculosis and Respiratory Diseases, Committee for the Revision of the Korean Guidelines for Tuberculosis, 2024). After liquid culture, we performed the MPT64 MTB-specific antigen test. In this study, we used the Biotline TB Ag MPT64 (Abbott Laboratories, Abbott Park, IL, USA) and performed the test according to the manufacturer's instructions.

This combination of AFB culture and MPT64 antigen testing served as the composite reference standard for this study, and its limitations

merit explicit statement. A positive liquid culture with a negative MPT64 antigen result was taken to indicate NTM growth; however, this approach does not constitute species-level identification of NTM. In addition, a small proportion of MTB isolates can yield false-negative MPT64 results for reasons unrelated to NTM, including specific MTB sublineages carrying deletions in the *mpt64* gene and pre-analytical or technical factors (Song et al., 2023). Because we applied this reference standard to all NTM-classified specimens in this study, the sensitivity and specificity estimates reported below for NTM should be interpreted as agreement with this composite, culture-based reference standard rather than as validated measures of species-confirmed diagnostic accuracy. We discuss this limitation, and its implications for our conclusions, further below.

2.4. Molecular diagnostic methods

The investigational method, GCdia, is a multiplex real-time PCR method that targets specific genes, including IS6110 and *MPB64* for MTB and a broad-range ITS gene region common to NTM species for NTM detection. Nucleic acid was extracted from sputum and cultured specimens according to the purification method of the QIAamp DSP DNA Mini Kit (QIAGEN, Hilden, Germany). The extracted DNA (15 μ L) was added to a PCR tube containing 15 μ L of MTBC/NTM PCR mixture, vortexed, and briefly centrifuged. We performed the assay using the CFX96Dx system (Bio-Rad Laboratories, Hercules, CA, USA). We defined positivity for TB and NTM as a cycle threshold (Ct) value ≤ 35 in the TB and NTM channels, and co-infection as positivity for both TB and NTM. A positive result in the internal control (IC) channel alone was considered negative.

The PCR gene target amplified by GCdia for MTB is referred to throughout this manuscript as *MPB64*, consistent with the assay's primer/probe design; this is distinct from the MPT64 secreted antigen detected by the comparator antigen test used in our reference standard (Bioline TB Ag MPT64, described above). Although *mpb64* and *mpt64* are sometimes used interchangeably in the diagnostic microbiology literature – both ultimately relate to the same *mpt64*/Rv1980c gene locus of MTBC, with “*MPB64*” historically denoting the *M. bovis*-derived designation of the PCR gene target and “*MPT64*” the *M. tuberculosis*-derived designation of the corresponding secreted protein/antigen – we retain each assay's original manufacturer nomenclature here (*MPB64* for the GCdia PCR target; *MPT64* for the antigen test) to avoid altering the names of the commercial products evaluated, and we have verified that each term is used consistently and only for its respective assay throughout the manuscript.

We tested a comparison control kit, Advansure, using an SLAN real-time quantitative PCR detection system (Shanghai Hongshi Medical Technology Co., Ltd.), according to the manufacturer's instructions. We analyzed three channels, FAM, HEX, and Cy5, to determine the presence of MTBC, NTM, or co-infection. We used the Ct value as the primary criterion for positive results. The TB channel (FAM) detects MTBC; a Ct value of <35 indicated a positive result. The NTM channel (HEX or Cy5) detected NTM; a Ct value of <40 indicated a positive result. The internal control (IC) channel confirmed PCR validity; a Ct value of <35 indicated a positive result.

For two discordant specimens with ambiguous results (D-08 and D-16; Table 7), we performed confirmatory testing targeting the *rpoB* gene. We confirmed MTB using primers TBC1 and TbcR5, and NTM using primers M5 and RM3. We sequenced PCR amplicons bidirectionally (forward and reverse) by Sanger sequencing and analyzed the resulting sequences by BLAST search and alignment against NCBI reference sequences to assign species identity.

2.5. Clinical sensitivity and specificity of the GCdia method

We compared the clinical sensitivity and specificity of the GCdia MTB/NTM Real-Time PCR method with those of the previously licensed

Advansure TB/NTM Real-Time PCR kit to assess concordance. This study was conducted from December 12, 2024, to December 31, 2025 (Ministry of Food and Drug Safety, Republic of Korea, 2025a). We distributed the specimens after approval from the MNTH Biobank Distribution Review Committee (Ministry of Food and Drug Safety, Republic of Korea, 2025b). We calculated the number of specimens using the formula $((\alpha + \beta)^2 p(1-p))/(p-p_0)^2$, based on reference literature and the licensed comparator's performance data (Ministry of Food and Drug Safety, 2024). Because specimens were dropped during evaluation, we calculated the total number of specimens as 660 by adding 5% to the minimum required number (Table 1).

Participants were selected if they met at least one of the following criteria: symptoms suggestive of pulmonary TB, suspected NTM infection, medical history of tuberculosis, or prior anti-tuberculosis medication use. We selected specimens based on valid reference methods that provided positive or negative TB results, in accordance with TB treatment guidelines (World Health Organization, 2026). Positive and negative specimens were selected if they met all of the following criteria: anonymized specimens, stored at -70°C or below, sputum and cultured specimens, specimens with a residual volume of ≥ 0.8 mL, and specimens with clear information related to the subject.

During the investigation, the MNTH specimen collection staff first verified the storage stability and suitability of the specimens against the selection criteria. A clinical performance test sample was selected from the population using random sampling (Ministry of Food and Drug Safety, 2024). To ensure anonymity, the specimen collection staff replaced personally identifiable information with the study number (Ministry of Food and Drug Safety, Republic of Korea, 2025c).

Next, Stat Trek's Random Number Generator generated random numbers for specimen randomization and assigned subject numbers (Stat Trek, 2025). Finally, the study number was removed to ensure blinding so that the analyst could not identify personally identifiable information or IVD evaluation results for the specimens. The specimen handover method was as follows: 660 sputum and culture-residual specimens from patients suspected of having MTBC or NTM infections were sorted by subject number and handed to an analyst.

The analyst conducted the experiments according to the protocols outlined in this study. The analyst recorded the test results in a case report form and reported them to the principal investigators. After the principal investigator's review, the analyst requested that the sample collection staff disclose the unblinded results; however, these unblinded results were excluded from the evaluation (Ministry of Food and Drug Safety, Republic of Korea, 2021).

2.6. Clinical performance evaluation and statistical analysis

We assessed clinical efficacy by determining whether the lower limits of the 95% confidence intervals for clinical sensitivity and specificity exceeded the lower limits of the 95% confidence intervals for the predefined clinical target sensitivity and specificity. We conducted statistical analyses using two-sided tests at the 5% significance level. We calculated the 95% confidence intervals for sensitivity and specificity using the Wald-Jeffrey method (Brown et al., 2001) (Table 2).

Cohen's kappa (κ) was used as a statistical method to measure

Table 1
Number of specimens used in this study.

Confirmed	Species	Specimen	Number	Subtotal	Total	
Positive	MTBC	Sputum	108	351	660	
		Cultured	108			
		Subtotal	216			
	NTM	Sputum	65	135		
		Cultured	70			
Negative	–	Subtotal	135	309		
		Sputum	148			
		Cultured	161			

Table 2

Clinical target values for sensitivity and specificity (95% CI).

Species	Specimen	Target Sensitivity (%)	Target Specificity (%)
MTBC	Sputum	≥ 99.00 (lower bound 95.46%)	≥ 99.00 (lower bound 94.76%)
	Culture	≥ 99.00 (lower bound 95.46%)	≥ 99.60 (lower bound 97.79%)
NTM	Sputum	≥ 95.26 (lower bound 85.65%)	≥ 99.00 (lower bound 94.76%)
	Culture	≥ 99.00 (lower bound 92.80%)	≥ 99.60 (lower bound 97.79%)

agreement, ranging from 0 to 1. $\kappa = 0$ indicated complete discrepancy between evaluations, whereas $\kappa = 1$ indicated complete agreement. We evaluated the reference value for Cohen's kappa agreement between the medical devices used for clinical performance testing and those used for controlled testing based on whether the Cohen's kappa (κ) index was ≥ 0.8 , with $0.8 < \kappa$ as the standard (Landis and Koch, 1977). We also summarized concordance between GCdia and Advansure using positive, negative, and overall percent agreement (PPA, NPA, and OPA, respectively).

Because GCdia and Advansure were both performed on the same specimens, differences in paired sensitivity and specificity between the two assays were evaluated using the exact McNemar test on discordant pairs, with 95% confidence intervals for the difference in proportions calculated using the standard method for correlated proportions; a two-sided p -value < 0.05 was considered statistically significant. All statistical calculations – including sensitivity, specificity, confidence intervals, PPA/NPA/OPA, Cohen's kappa, and the exact McNemar test – were performed using Microsoft Excel (Microsoft Corporation, Redmond, WA, USA).

3. Results

We conducted a clinical performance test on 660 residual samples collected during routine diagnosis. We tested patients suspected of having MTB or NTM infections using the GCdia method and an Advansure control device (Table 3).

We calculated the sensitivity and specificity of GCdia and Advansure using the composite reference standard (AFB culture with MPT64 antigen differentiation; see Materials and Methods) in 660 clinical specimens. For sputum specimens, the sensitivity and specificity for detecting MTB were 100.0% (108/108; 95% CI 97.71–100) and 99.32% (147/148; 95% CI 96.89–99.93) for GCdia, and 95.37% (103/108; 95% CI 90.15–98.21) and 98.65% (146/148; 95% CI 95.74–99.72) for Advansure, respectively. The difference in paired sensitivity was 4.63 percentage points (95% CI 0.67 to 8.59; exact McNemar $p = 0.063$), a trend toward higher sensitivity for GCdia that did not reach statistical

significance.

The sensitivity and specificity for detecting NTM were 100.0% (65/65; 95% CI 96.22–100) for GCdia, 89.23% (58/65; 95% CI 80.02–95.05) for Advansure, and 100.0% (148/148; 95% CI 98.32–100) for both GCdia and Advansure. This 10.77-percentage-point difference in sensitivity was statistically significant (95% CI 3.23 to 18.31; exact McNemar $p = 0.016$).

When cultured specimens rather than sputum were used, the sensitivity and specificity for detecting MTB were 99.07% (107/108; 95% CI 95.75–99.90) for GCdia and 100.0% (108/108; 95% CI 97.71–100) for Advansure, with a specificity of 100.0% (161/161; 95% CI 98.45–100) for both. The one-specimen difference in culture MTB sensitivity was not statistically significant (95% CI -2.73 to 0.88 ; exact McNemar $p = 1.000$). The sensitivity and specificity for detecting NTM were 100.0% (70/70; 95% CI 96.49–100) for both GCdia and Advansure, with a specificity of 100.0% (161/161; 95% CI 98.45–100) for both (Table 3); no discordant pairs were observed for culture NTM.

For all four target-and-specimen combinations, the lower bound of the 95% confidence interval for sensitivity and specificity met or exceeded the predefined clinical performance targets (Table 2). Overall, the GCdia method demonstrated clinical effectiveness, with sensitivity and specificity for detecting MTB and NTM in both sputum and cultured specimens that were comparable to, or, for NTM detection in sputum, significantly higher than Advansure when compared with the composite reference standard.

We calculated the concordance rate between the test method (GCdia PCR) and the previously licensed comparator (Advansure PCR) by comparing results on 660 clinical specimens, stratified by sputum and culture specimen types (Table 4).

For sputum specimens, the positive percent agreement (PPA) for MTB detection was 98.1% (95% CI 94.03–99.6%), the negative percent agreement (NPA) was 96.02% (95% CI 92.34–98.4%), and the overall

Table 4

Concordance of the GCdia PCR and Advansure PCR assays for detection of MTB and NTM in sputum specimens.

		GCdia: MTB	GCdia: NTM	GCdia: Negative	Total (%)
Advansure	MTB	103	0	2	105 (32.71)
	NTM	0	58	0	58 (18.07)
	Negative	6	7	145	158 (49.22)
	Total	109 (33.97)	65 (20.24)	147 (45.79)	321 (100.0)

Rows: Advansure PCR result. Columns: GCdia PCR result. $n = 321$ sputum specimens.

Table 3

Comparison of the sensitivity and specificity of the GCdia PCR and Advansure PCR assays for the detection of MTB and NTM using sputum and culture specimens.

Specimen	Target	Reference (No.) (Pos/ Neg)	Assay	True positive	False positive	Sensitivity %, (95% CI)	Specificity %, (95% CI)	Sens. diff., % (95% CI)*	McNemar p^*
Sputum	MTB	108/148	GCdia	108/108	1/148	100.0 (97.71–100)	99.32 (96.89–99.93)	4.63 (0.67 to 8.59)	0.063
			Advansure	103/108	2/148	95.37 (90.15–98.21)	98.65 (95.74–99.72)		
	NTM	65/148	GCdia	65/65	0/148	100.0 (96.22–100)	100.0 (98.32–100)	10.77 (3.23 to 18.31)	0.016
			Advansure	58/65	0/148	89.23 (80.02–95.05)	100.0 (98.32–100)		
Culture	MTB	108/161	GCdia	107/108	0/161	99.07 (95.75–99.90)	100.0 (98.45–100)	-0.93 (-2.73 to 0.88)	1.000
			Advansure	108/108	0/161	100.0 (97.71–100)	100.0 (98.45–100)		
	NTM	70/161	GCdia	70/70	0/161	100.0 (96.49–100)	100.0 (98.45–100)	0 (no discordant pairs)	NA
			Advansure	70/70	0/161	100.0 (96.49–100)	100.0 (98.45–100)		

Reference standard: composite reference standard (AFB culture with MPT64 antigen differentiation; see Materials and Methods for limitations). We commonly used negative specimens (148 for sputum and 161 for culture) for both MTB and NTM analyses. 95% CIs were calculated by the Wald–Jeffrey (Jeffreys) method. *Sensitivity difference (GCdia – Advansure) and exact McNemar p -value for paired discordant results between the two assays on identical specimens; specificity differences were not statistically significant for any target-specimen combination (exact McNemar $p = 1.000$ or not applicable) and are not tabulated separately.

Table 5

Concordance of the GCdia PCR and Advansure PCR assays for detection of MTB and NTM in culture specimens.

		GCdia: MTB	GCdia: NTM	GCdia: Negative	Total (%)
Advansure	MTB	107	1	0	108 (31.86)
	NTM	0	70	0	70 (20.65)
	Negative	0	0	161	161 (47.49)
	Total	107 (31.56)	71 (20.94)	161 (47.49)	339 (100.0)

Rows: Advansure PCR result. Columns: GCdia PCR result. n = 339 culture specimens.

percent agreement (OPA) was 96.96% (95% CI 94.34–98.55%). The PPA for NTM detection was 100% (95% CI 95.78–100%), the NPA was 95.57% (95% CI 91.5–98%), and the OPA was 96.76% (95% CI 93.74–98.54%).

For culture specimens, the PPA for MTB detection was 99.07% (95% CI 95.75–99.9%), the NPA was 100% (95% CI 98.45–100%), and the OPA was 99.63% (95% CI 98.28–99.96%). The PPA for NTM detection was 100% (95% CI 96.49–100%), the NPA was 100% (95% CI 98.45–100%), and the OPA was 100% (95% CI 98.92–100%). In all cases, the Cohen's kappa (κ) value was ≥ 0.9 , indicating a very high degree of agreement (Table 6).

A comparison between the GCdia and Advansure methods showed high concordance, with kappa (κ) values all >0.9 , indicating excellent agreement. A total of 16 specimens showed a discordant result between GCdia and Advansure: 15 among sputum specimens and 1 among culture specimens. In sputum, 2 specimens were Advansure-positive for MTB but GCdia-negative; 6 specimens were GCdia-positive for MTB, but Advansure-negative; and 7 specimens were GCdia-positive for NTM but Advansure-negative. In culture, 1 specimen was classified as MTB-positive by the reference method and by Advansure but was GCdia-positive for NTM. A summary of these 16 discordant specimens, including individual Ct values and confirmatory testing where available, is provided in Table 7. Given the overall performance, the GCdia PCR method showed clinical utility comparable to, or for NTM detection in sputum significantly higher than, the licensed comparator assay.

4. Discussion

Tuberculosis (TB) is a respiratory infectious disease, and early detection and treatment are essential to prevent its spread. However, its diverse clinical manifestations can make diagnosis challenging, and distinguishing it from non-tuberculous mycobacterial (NTM) disease can be difficult in the early stages. NTM infections are a significant concern, comparable to TB, as patients with NTM also face challenges such as prolonged treatment periods, adverse drug effects, and high recurrence rates. According to a study at the Duke University Health System (DUHS) in the United States, patients with NTM pulmonary disease (NTM-PD) had an approximately 4–8.6-year reduction in life

expectancy, and their five-year mortality rate was roughly four times higher than that of the general population (Mourad et al., 2021).

For diagnosing both MTB and NTM infections, the sensitivity of acid-fast bacilli (AFB) microscopy is approximately 25–80% lower than that of culture methods, and it cannot distinguish between MTB and NTM (Korean Academy of Tuberculosis and Respiratory Disease, 2024). Furthermore, AFB culture is disadvantageous because it is time-consuming to obtain results. Consequently, molecular diagnostic methods, particularly real-time PCR, which offer rapid testing and high sensitivity, are widely used to diagnose MTB and NTM infections (Drosten et al., 2003).

4.1. Clinical performance evaluation of the GCdia MTB/NTM real-time PCR method

This study validated the clinical sensitivity and specificity of the GCdia MTB/NTM Real-Time PCR method using sputum and culture samples from patients with suspected pulmonary TB and NTM infection. The clinical performance test demonstrated that the lower limit of the 95% confidence interval for the method's sensitivity and specificity met the predefined clinical standards, indicating clinically meaningful diagnostic performance. Additionally, in an evaluation of its consistency with the Advansure PCR kit, an already licensed and commercialized product in Korea, the kappa (κ) values were confirmed to be over 0.9, and paired comparison using the exact McNemar test showed a statistically significant advantage for GCdia in NTM detection from sputum ($p = 0.016$); differences for MTB in sputum and for both targets in culture specimens were not statistically significant.

The GCdia PCR method targets a broad-range ITS gene region rather than relying on species-specific primers, and in this study all NTM-culture specimens meeting the composite reference standard described above were consistently classified as NTM-positive by GCdia, supporting its utility as a broad-range NTM screening assay. While the existing Advansure PCR kit determines TB infection or TB/NTM co-infection using both the cutoff value and the difference in Ct values between TB and NTM, the GCdia PCR method makes its determination solely on the cutoff value; this simpler decision rule may offer more intuitive result interpretation than the Advansure PCR kit, although a dedicated usability evaluation was not performed in this study.

4.2. Study limitations and future research

The most important limitation of this study concerns the reference standard used to classify NTM. As described in Materials and Methods, NTM-culture specimens were considered NTM if the MPT64 antigen test result was negative after liquid culture; this does not constitute a definitive species-level diagnosis, and a small proportion of MTB isolates can yield false-negative MPT64 results for reasons unrelated to NTM (Song et al., 2023). Because this reference standard was used for all NTM-classified specimens in this study, the sensitivity and specificity estimates reported here for NTM should be interpreted as estimates of agreement with a composite, culture-based reference standard rather than as validated measures of species-confirmed diagnostic accuracy, and we have moderated our conclusions throughout this manuscript

Table 6

Percent agreement (PPA, NPA, OPA) of the GCdia PCR and Advansure PCR assays for detection of MTB and NTM in sputum and culture specimens.

Specimen	Species	Percent Agreement (%), (95% CI)	Kappa (κ)	Specimen	Species	Percent Agreement (%), (95% CI)	Kappa (κ)
Sputum	MTB	PPA: 98.1 (94.03–99.6)	0.937	Culture	MTB	PPA: 99.07 (95.75–99.9)	0.992
		NPA: 96.02 (92.34–98.4)				NPA: 100 (98.45–100)	
		OPA: 96.96 (94.34–98.55)				OPA: 99.63 (98.28–99.96)	
	NTM	PPA: 100 (95.78–100)	NTM		PPA: 100 (96.49–100)	1.000	
		NPA: 95.57 (91.5–98)			NPA: 100 (98.45–100)		
		OPA: 96.76 (93.74–98.54)			OPA: 100 (98.92–100)		

PPA, positive percent agreement (proportion of Advansure-positive specimens also positive by GCdia, and vice versa, for the indicated species); NPA, negative percent agreement; OPA, overall percent agreement. All figures compare GCdia against Advansure directly (not against the reference standard).

Table 7
Summary of discordant specimens between the GCdia and Advansure assays.

ID	Specimen	Reference Result	GCdia Result	Advansure Result	GCdia Ct (FAM = NTM/HEX = MTB/Cy5 = IC)	Advansure Ct (FAM = MTB/HEX = NTM/Cy5 = IC)	Confirmatory Testing / Notes
D-01	Sputum	Negative	Negative	MTB	0.00/0.00/26.25	27.85/32.72/27.59	Not performed
D-02	Sputum	Negative	Negative	MTB	0.00/0.00/26.11	8.04/8.11/27.48	Not performed
D-03	Sputum	MTB	MTB	Negative	0.00/32.65/26.47	No Ct/No Ct/27.95	Not performed
D-04	Sputum	MTB	MTB	Negative	0.00/32.07/26.66	No Ct/No Ct/27.47	Not performed
D-05	Sputum	MTB	MTB	Negative	0.00/34.68/25.70	No Ct/No Ct/27.81	Not performed
D-06	Sputum	MTB	MTB	Negative	0.00/31.68/25.88	No Ct/No Ct/27.43	Not performed
D-07	Sputum	MTB	MTB	Negative	0.00/34.10/26.22	No Ct/No Ct/27.80	Not performed
D-08	Sputum	Negative	MTB	Negative	0.00/32.12/25.10	No Ct/No Ct/27.93	Sanger Sequencing: inconclusive
D-09	Sputum	NTM	NTM	Negative	32.91/0.00/25.11	No Ct/No Ct/27.57	Not performed
D-10	Sputum	NTM	NTM	Negative	30.28/0.00/25.17	No Ct/No Ct/27.57	Not performed
D-11	Sputum	NTM	NTM	Negative	32.53/0.00/24.80	No Ct/No Ct/27.38	Not performed
D-12	Sputum	NTM	NTM	Negative	27.99/0.00/25.24	No Ct/No Ct/28.56	Not performed
D-13	Sputum	NTM	NTM	Negative	22.31/0.00/25.57	No Ct/No Ct/27.22	Not performed
D-14	Sputum	NTM	NTM	Negative	30.94/0.00/25.19	No Ct/No Ct/27.32	Not performed
D-15	Sputum	NTM	NTM	Negative	29.70/0.00/25.01	No Ct/No Ct/27.14	Not performed
D-16	Culture	MTB	NTM	MTB	24.90/0.00/25.18	20.23/26.33/27.35	Sanger Sequencing: MTB

Summary of all 16 specimens with a discordant result between GCdia and Advansure (see Results). “Reference Result,” “GCdia Result,” and “Advansure Result” columns are derived from the categorical concordance data in Tables 4 and 5. We show individual Ct values for each channel (FAM/HEX/Cy5) in the GCdia and Advansure assays. Confirmatory Sanger sequencing was performed for two specimens with ambiguous or discrepant results (D-08, D-16; see Materials and Methods for the confirmatory testing protocol); confirmatory testing was not performed for the remaining 14 discordant specimens. Channel assignments differ between assays: for GCdia, FAM corresponds to the NTM target, HEX to the MTB target, and Cy5 to the internal control (IC); for Advansure, FAM corresponds to the MTB (TB) target, HEX to the NTM target, and Cy5 to the IC. The unusually low Advansure FAM/HEX Ct values for specimen D-02 (8.04/8.11) were verified against the original laboratory record and reflect genuine measured values; the specific cause of this atypical result was not established.

accordingly.

This study has several additional limitations. We compared GCdia against a single licensed comparator assay (Advansure); comparing it with additional comparator assays would help generalize these findings. This was also a single-center study using banked residual specimens from one institution, which may limit the generalizability of these results to other populations and laboratory settings. Finally, although the paired design allowed formal statistical comparison of GCdia and Advansure, the number of discordant results was small for several target-and-specimen combinations (Table 7), limiting the precision of the corresponding confidence intervals. Future research incorporating molecular or sequencing-based NTM species identification, additional comparator assays, and multi-center enrollment is needed to confirm and extend these findings across a wider variety of NTM species and clinical settings.

4.3. Methodological considerations for domestic in vitro diagnostic medical device approval

This clinical performance test was conducted in accordance with the Korean regulations and guidelines for in vitro diagnostic medical devices. Validation of the clinical effectiveness of the GCdia method also adhered to the standards required for domestic IVD approval. The MFDS highlighted the following precautions during actual clinical performance testing. These methodological considerations are summarized to assist groups seeking future approval for TB/NTM molecular diagnostic methods in Korea.

4.3.1. Clarity in test subject selection criteria

The criteria for selecting patients with suspected MTB and NTM infections should be based on clinical standards as stipulated in the Korean tuberculosis treatment guidelines. The criteria for suspected tuberculosis must include specific clinical information, such as chest X-ray findings, AFB smear results, cough, and sputum, and the negative control group should also be clearly defined (Ministry of Food and Drug Safety, Republic of Korea, 2021).

4.3.2. Clarification of sample random sampling and randomization procedures

Ambiguous procedures for sample collection and randomization may compromise the test's reliability. Therefore, the research plan should clearly describe the extraction process, including random sampling and the use of random number generators (Ministry of Food and Drug Safety, Republic of Korea, 2021).

4.3.3. Demonstrating the stability of long-term stored samples

When using long-term-stored samples for clinical performance tests, a clear record of the storage conditions, including temperature, duration, and the number of freeze-thaw cycles, is required. Secure data, such as internal quality control records, to demonstrate that these samples remain stable (Ministry of Food and Drug Safety, Republic of Korea, 2021).

4.3.4. Compliance with the research plan and institutional standard operating procedure (SOP)

Clinical performance tests must be conducted in accordance with the relevant institution's standard operating procedures (SOP) or an official test plan. The absence of a plan or non-compliance with the SOP may result in rejection of approval. Therefore, during the research plan writing stage, the institution's SOPs or instructions for using the devices in clinical performance tests must be clearly defined (Ministry of Food and Drug Safety, Republic of Korea, 2021).

5. Conclusion

This study evaluated the clinical efficacy and reliability of the

GCdiaMTB/NTM Real-Time PCR method. The method's sensitivity and specificity, along with its strong concordance with a licensed product, suggest it is a valuable diagnostic tool for rapid diagnosis of TB and NTM infections. Because the reference standard used for NTM classification does not include species-level confirmation, these findings should be interpreted as evidence of agreement with a composite reference standard for NTM detection rather than validated species-level diagnostic accuracy; confirmation using molecular or sequencing-based species identification, and comparison against additional comparator assays in multi-center studies, are warranted before broader clinical claims can be made. Insights from the Korean IVD approval process provide a practical methodological guide for future research and commercial development of molecular diagnostic methods in this field.

Regulatory note

The clinical performance testing described in this manuscript was conducted by the Clinical Research Center of Masan National Tuberculosis Hospital (MNTH), a nationally designated in vitro diagnostic medical device clinical performance testing institution (IVD Notice No. 79, July 1, 2021) under the Korean Ministry of Food and Drug Safety (MFDS). MNTH served as an independent evaluation center; the study was neither commissioned nor funded by the kit manufacturers, nor by MycoDX Co., Ltd. Following completion of the study, and independently of this manuscript, the GCdia kit received product approval from the Korean MFDS (In Vitro Diagnostic Device License No. 25-660). The authors had no role in, and derive no financial benefit from, this regulatory approval.

CRediT authorship contribution statement

Seo Yong Choe: Writing – original draft, Methodology, Investigation. **Da-Gyum Lee:** Investigation, Data curation. **Jin-A Park:** Resources, Project administration, Methodology, Data curation. **Munki Yoo:** Methodology, Investigation. **Jaehyeon Kang:** Validation, Resources, Methodology. **Tak Gyeom Lee:** Resources, Methodology, Data curation. **JeongA Kim:** Resources, Methodology, Formal analysis. **Dabin Jeong:** Methodology, Data curation. **Soozy Kim:** Resources, Methodology. **Younghoon Jang:** Visualization, Validation, Data curation. **Heedoo Lee:** Validation, Supervision, Conceptualization. **Jichan Jang:** Writing – review & editing, Visualization, Data curation, Conceptualization. **Sungweon Ryoo:** Writing – review & editing, Supervision, Conceptualization.

Ethics statement

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki. The Institutional Review Board of Masan National Tuberculosis Hospital approved the study protocol (approval number: IRB-398837MD-2024-E01). All specimens were obtained from the MNTH BioBank under a separately authorized biobank protocol (approval number: IRB-07-N03). Informed consent was waived by the IRB owing to the retrospective, anonymized nature of residual specimen use.

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Declaration of competing interest

The corresponding author (Sungweon Ryoo) and two co-authors (Jin-A Park and JeongA Kim) have dual affiliations with MycoDx Co., Ltd., a for-profit biotechnology company engaged in tuberculosis molecular diagnostics, in addition to their academic/institutional affiliations. This affiliation is disclosed here in the interest of full transparency. MycoDx Co., Ltd. did not manufacture, distribute, fund, or otherwise have any commercial interest in either of the diagnostic products evaluated in this study – the investigational GCdiaMTB/NTM Real-Time PCR Kit (Genes Lab, Gyeonggi-do, Korea) or the comparator AdvansureTB/NTM Real-Time PCR kit (Invitrogen, Osong, Korea) – and had no role in the design, conduct, analysis, interpretation, or reporting of this study.

Genes Laboratories Co., Ltd., the manufacturer of the investigational GCdiaMTB/NTM Real-Time PCR Kit, provided the test kits used in this study free of charge and trained study staff on the assay protocol before testing. Genes Laboratories Co., Ltd. had no role in specimen selection, randomization, blinding, data analysis, interpretation of results, or the decision to submit this manuscript for publication, and no author received personal payment, consulting fees, honoraria, or any other financial benefit from Genes Laboratories Co., Ltd. This free provision of test kits and protocol training is disclosed here as an in-kind contribution; it did not extend to any role in the scientific conduct or reporting of the study.

Aside from these disclosed relationships, the authors declare no other known competing financial interests or personal relationships that could have influenced the work reported in this paper. Specifically, none of the authors has: received financial compensation, honoraria, consulting fees, or speaker fees from either kit manufacturer (Genes Lab or Invitrogen) related to the subject matter of this manuscript; held or applied for patents related to the diagnostic methods, kits, or technologies described herein; held stock or equity interests in Genes Lab, Invitrogen, or any other company manufacturing or distributing the products evaluated in this study; or received research funding directly from either kit manufacturer for the conduct of this study.

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Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request, subject to applicable data privacy and institutional regulations.

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